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3D Coupled Modelling of a Series-Connected Bifilar Copper-Stabilised REBCO Resistive Superconducting Fault Current Limiter

August 2026

A thesis submitted in partial fulfilment of the requirements for the degree of
MASTER OF SCIENCE IN ELECTRONICS AND ELECTRICAL ENGINEERING

Abstract

Distributed synchronous generation raises the fault current at the point of common coupling on a low-voltage rural feeder. A three-dimensional model coupling the electromagnetic field, heat transfer and the external circuit is developed for a series-connected bifilar copper-stabilised REBCO resistive superconducting fault current limiter. A 50 mm segment, discretised with 119,184 elements and 554,999 degrees of freedom, represents the full device. For the 50 ms base case, the first fault-current peak is limited from 2.93 kA in the zero-impedance reference to 456.1 A, corresponding to an 84.4% reduction. The terminal voltage reaches approximately 309 V, while the superconducting-layer hot spot rises to approximately 81.9 K. Clearing times of 20 ms or longer leave the first current peak unchanged but increase the subsequent energy deposition and temperature rise. At the end of the solution window, the electrical-port input differs from the Joule heat by 0.81%. The limiter therefore suppresses the first fault peak before mechanical interruption, whereas the opening instant and liquid-nitrogen heat transfer determine the subsequent thermal response. An otherwise identical uniform-Jc control raises the first current peak by 0.31% and lowers the superconducting-layer extremum by approximately 1 K, indicating that the prescribed periodic weak-zone pattern primarily affects local heating in this operating case.

Acknowledgements

I thank my supervisor for proposing this topic and for the guidance that kept the modelling work on course throughout the project. I am grateful as well to my fellow students in the School of Engineering, whose questions on several occasions sent me back to check a result that turned out to be worth checking.

Principal symbols

The following tables group the principal symbols used in the text and figures into basic constants, electromagnetic quantities, heat-transfer quantities and abbreviations. Other quantities are defined at first use.

Basic and model constants	Meaning	Unit
μ_0	Permeability of free space	$H \cdot m^{-1}$
T_c	REBCO critical temperature (model reference)	K
E_c	Critical-current criterion electric field	$V \cdot m^{-1}$
B_0	Kim-model characteristic field	T

Electromagnetic quantities	Meaning	Unit
$\boldsymbol{H}$	Magnetic field strength vector (solved variable)	$A \cdot m^{-1}$
$\boldsymbol{B}$	Magnetic flux density vector	T
$B_\perp$, $B_\parallel$	Flux-density components perpendicular/parallel to the tape face	T
$\boldsymbol{E}$	Electric field strength vector	$V \cdot m^{-1}$
$\boldsymbol{J}$	Current density vector	$A \cdot m^{-2}$
$J_c(T, B)$	Critical current density (temperature- and field-dependent)	$A \cdot m^{-2}$
n	E-J power-law index	N/A
k_w	Weak-zone critical-current reduction factor	N/A
ρ	Resistivity	$\Omega \cdot m$
i	Instantaneous loop current	A
u_{dev}	Limiter terminal voltage	V
R_s, L_s	Source-side equivalent resistance and inductance	Ω, H
W_{mag}	Stored magnetic energy	J

Heat-transfer quantities	Meaning	Unit
T	Temperature	K
T_b	Liquid-nitrogen bath temperature	K
ρ_m	Mass density	$kg \cdot m^{-3}$
c_p	Specific heat capacity at constant pressure	$J \cdot kg^{-1} \cdot K^{-1}$

Heat-transfer quantities	Meaning	Unit
λ	Thermal conductivity	$W \cdot m^{-1} \cdot K^{-1}$
Q_{v}	Volumetric Joule heat source density	$W \cdot m^{-3}$
q	Wall heat flux	$W \cdot m^{-2}$
$h_{LN_2}(\Delta T)$	Liquid-nitrogen boiling heat-transfer coefficient	$W \cdot m^{-2} \cdot K^{-1}$
ΔT	Wall superheat (wall temperature minus bath temperature)	K

Abbreviation	Meaning
SFCL	Superconducting fault current limiter
REBCO	Rare-earth barium copper oxide coated conductor ($RE\text{-}Ba_2Cu_3O_{7-\delta}$)
HTS	High-temperature superconductor
CHP	Combined heat and power unit
LN_2	Liquid nitrogen
KVL	Kirchhoff's voltage law
FEM	Finite element method
CHF	Critical heat flux (boiling crisis)
RMS	Root mean square

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1. Introduction

Distributed generation alters the magnitude and distribution of fault current in a distribution network. During the sub-transient stage, a directly connected rotating machine may deliver several times its rated current, with the contribution decaying from 400% to 200% of rating over a few power-frequency cycles. Such machines can therefore dominate the increase in short-circuit level at the point of common coupling (Keller and Kroposki, 2010). Photovoltaic and modern wind-turbine converters are instead constrained by their control systems; current-limiting and protection functions act at approximately 1–2 times rated current, while UL 1741 tests have recorded peaks of 2–5 times rating lasting about 1–4.25 ms (Keller and Kroposki, 2010). Such brief contributions chiefly affect the sensitivity and reach of overcurrent protection in inverter-dominated networks (Slane and Mate, 2026). The case considered here is a synchronous combined heat and power unit connected at low penetration to a rural feeder.

Uprating switchgear raises refurbishment cost, bus splitting reduces operational flexibility, and a series reactor imposes standing impedance, loss and voltage offset on the healthy network. A resistive SFCL stays in its low-resistance state at rated current; once the fault current exceeds the critical value, the REBCO conductor quenches ahead of any mechanical switching action and builds a dynamic resistance that limits the first current peak.

The limiting response develops over milliseconds. Joule heat generated in the superconducting layer passes through the stabiliser to the liquid nitrogen. As the conductor warms, the critical current density falls and the local resistance rises, changing both the loop current and the power deposition. The electromagnetic field, temperature field and external circuit must therefore be solved as a coupled transient.

A three-dimensional field-circuit model is used to calculate the limited current, local temperature rise and post-interruption cooling, and to determine how the clearing time modifies the thermal response. The effects of the liquid-nitrogen boundary condition and thin-layer discretisation are assessed with the numerical results.

1.1 Aim and tasks

The model combines the three-dimensional electrothermal response of the device with the external-circuit transient. It follows the limited current, terminal voltage, temperature and equivalent resistance through the fault and subsequent thermal relaxation. Clearing time is the principal operating parameter, and a uniform-Jc control isolates the effect of the prescribed weak-zone pattern.

1.2 Object of study and operating case

The computational domain contains a 50 mm representative conductor segment. The series-array relations in Chapter 3 map segment quantities to the full device. The cryostat, mechanical supports and current leads lie outside the domain, while lumped elements represent the system-side switchgear.

The reference network is a 400 V rural feeder supplied by a 50 kVA pole-mounted distribution transformer. A short-circuit impedance of about 5% gives a prospective short-circuit capacity of approximately 1 MVA at the point of common coupling. A 35 kW biogas combined heat and power unit is connected through a directly coupled synchronous generator rated at 35 kVA and 51 A rms. The SFCL is installed in series at the generator connection to limit its contribution to the fault current. At unity power factor, the rated apparent and active powers are numerically equal.

The 50 kVA transformer defines the source impedance of the rural low-voltage feeder. The 35 kW generator lies within the 15–100 kW electrical range reported for farm-scale anaerobic digestion plants (O'Connor et al., 2021) and is comparable with the 41–45 kW commercial CHP units used in a UK mixed-farm study (Shorrocks, Wang and Huang, 2025). Its fault contribution is concentrated in the first tens of milliseconds after inception, matching the transient interval analysed here.

1.3 Structure of the thesis

The analysis begins with the coated-conductor stack, limiter topology and heat-transfer evidence needed to describe quench propagation in liquid nitrogen (Chapter 2). These elements are then assembled in a three-dimensional model that solves the electromagnetic field, solid heat transfer and external circuit simultaneously (Chapter 3).

The resulting electrical and thermal transients, including their sensitivity to the prescribed weak-zone pattern, are examined through the fault current, terminal voltage, local field variables, energy balance, rated-current loss and magnetic-field redistribution (Chapter 4). Their principal implications are summarised in Chapter 5, while Chapter 6 addresses geometry-specific heat-transfer calibration, winding-dependent loss, independent validation and loaded recovery.

2. Literature Review

2.1 Coated conductors for power applications

Second-generation high-temperature superconducting tape is generally manufactured as a coated conductor. The metal substrate provides mechanical support and a texture template, buffer layers prevent interdiffusion while transmitting the biaxial texture, and a REBCO film about 1 μm thick carries the superconducting current. Silver provides a low-resistance contact and protects the REBCO. In power devices, an additional copper stabiliser receives current after quench and spreads the resulting heat along the conductor.

The copper thickness determines the capacity for current sharing and longitudinal heat spreading after quench. Increasing it moderates the temperature rise but adds material, device volume and normal-state loss. The stack adopted in the model is specified in §3.3.

2.2 Limiter topologies

SFCLs are commonly classified as resistive, inductive or hybrid. A resistive device develops its limiting impedance within the superconducting element. Inductive designs act through magnetic coupling, with saturated-core variants requiring a DC bias winding. Hybrid arrangements use a superconducting element to initiate commutation into a parallel limiting branch.

The analysis concerns a resistive SFCL, in which the coated-conductor stack provides the limiting impedance. Its winding connection also determines the net loop flux linkage, equivalent inductance and AC loss, allowing the electrical and thermal consequences of counter-winding to be evaluated within the same model.

2.3 Experimental studies of resistive SFCLs

Experimental studies of resistive SFCLs generally apply an overcurrent of prescribed duration to a coil section or coated-conductor stack and record the limited current, terminal voltage, temperature and recovery. Song et al. (2021a) combined tests and simulation for a helical limiter; a companion study distinguished recovery under sustained load current from recovery under near-unloaded conditions (Song et al., 2021b). Chen et al. (2022) treated fault duration as an independent parameter in tests of a coated-conductor limiter, whereas Rocha et al. (2021) specified the impulse length in power-frequency cycles. Sheng et al. (2016)

examined the repeated overcurrent withstand of coated conductors immersed in liquid nitrogen.

These experiments provide useful reference points for recovery criteria and fault duration, although quantitative comparisons require allowance for differences in topology, voltage class and conductor specification.

2.4 Numerical formulations for coated-conductor devices

Coated-conductor devices are modelled principally with the H-formulation, the T-A formulation, and integral or lumped-parameter methods. The H-formulation accommodates the strongly nonlinear resistivity of three-dimensional stacks and is adopted for the electromagnetic sub-model (Shen, Grilli and Coombs, 2020). Because power-law extrapolation can overstate resistance growth far above the critical current (Riva et al., 2020), §3.2 specifies the overcritical treatment used in the calculation. Colangelo and Dutoit (2012) examined how longitudinal critical-current inhomogeneity affects partial quench and power dissipation in resistive SFCLs.

2.5 Fault-time parameters: definitions and scope

Three-time quantities are distinguished in this study. Experimental fault duration extends from application to removal of the short-circuit source. Fault clearing time extends from inception to complete interruption of the loop current. Rated breaking time is measured from trip-coil energisation to arc extinction and excludes protection and command delays.

Tixador (2023) identifies 20 ms as a fast-clearing lower limit and 50 ms as a design allowance. The latter is used as the base case, with shorter and longer values included in the parameter sweep. Published device tests often apply faults for approximately 100 ms (Song et al., 2021a, 2021b; Chen et al., 2022), but those durations use different measurement boundaries and are not model inputs.

2.6 Liquid-nitrogen boiling heat transfer

Studies of liquid-nitrogen cooling for resistive SFCLs address three related regimes: steady pool boiling, rapid transient heating and visually resolved bubble dynamics.

Steady pool-boiling measurements define the nucleate-boiling branch, critical heat flux and the effects of heater orientation and liquid subcooling (Ahmad et al., 2025; Merte, 1970).

Because these correlations assume developed flow, they do not describe a fault pulse lasting only tens of milliseconds.

Balakin et al. (2017) and Delov et al. (2020) interpret the boiling crisis under rapid heating through the competing timescales of boiling inception and natural convection. Their results also show that critical heat flux depends on heater dimensions as well as fluid state, with plate and strip heaters occupying the lower threshold range. Fine-wire data are consequently unsuitable as a direct transient boundary condition for coated conductor.

Visualisation experiments with a closer geometric resemblance show post-quench bubbles becoming trapped between the tape and its support (Zhi et al., 2020). If liquid replenishment through a narrow channel cannot match vapour generation, the microlayer is depleted and rewetting slows (Luo et al., 2022a, 2022b).

Rubeli et al. (2015) compared the measured conductor temperature with an adiabatic thermal model and used the difference to estimate heat transfer to the liquid-nitrogen bath. Sheng et al. (2016) observed a transient decrease in resistance and temperature during the final part of the overcurrent pulse as the boiling regime changed. Neither study, however, provides a critical heat flux calibration for the present geometry.

No published dataset simultaneously covers copper-stabilised coated conductors, vertical narrow gaps and heating over tens of milliseconds. The proposed calibration therefore targets this combination of geometry and timescale.

2.7 Research gap

The literature separately addresses coated-conductor materials, SFCL tests, numerical methods and liquid-nitrogen boiling. A unified treatment of the coupled transient response of a series-connected counter-wound stack is still lacking, including its sensitivity to prescribed longitudinal critical-current non-uniformity.

3. Methods

3.1 Overview and coupling structure

The electromagnetic, thermal and circuit equations are solved simultaneously. The field solution supplies electric field and current density to the thermal model through the Joule heat source. Temperature then updates the critical current density and material resistivities. At the circuit interface, the external network provides the transport current and receives the terminal voltage integrated from the field solution.

Figure 3.1 identifies the quantities exchanged between the electromagnetic, thermal and circuit domains.

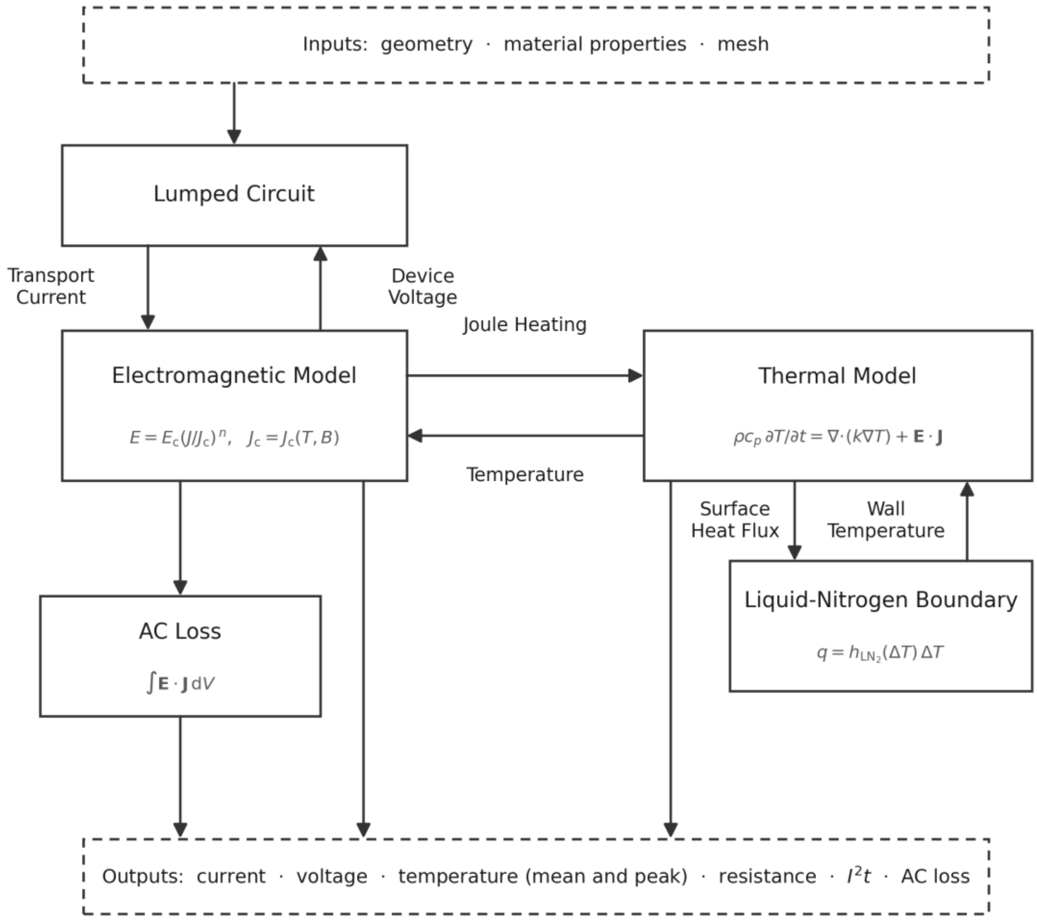

Figure 3.1 Coupling structure of the simulation.

3.2 Governing equations

3.2.1 Electromagnetic field

The electromagnetic problem is written in the H-formulation. It admits the strongly nonlinear superconducting resistivity directly and avoids an auxiliary scalar potential within the conductors, making it suitable for a three-dimensional tape stack.

The governing equations follow from Maxwell's equations:

$$\nabla \times \boldsymbol{E} = -\mu_0 \frac{\partial \boldsymbol{H}}{\partial t}$$

$$\boldsymbol{J} = \nabla \times \boldsymbol{H}$$

Here, $\boldsymbol{E}$ denotes electric field strength, $\boldsymbol{H}$ magnetic field strength, $\boldsymbol{J}$ current density and μ_0 the permeability of free space.

3.2.2 Superconducting-layer constitutive law

A resistive SFCL presents negligible impedance below its critical current. Once the fault current exceeds that threshold, dissipation in the superconducting layer produces the resistance required for limitation without an external trigger (Noe and Steurer, 2007; Tixador, 2023). The constitutive law therefore determines both the onset and the rate of impedance growth.

The superconducting layer is described by an anisotropic power law relating E to J. The electric field is parallel to the current density and has the magnitude

$$\boldsymbol{E} = E_{\mathrm{c}} \left(\frac{|\boldsymbol{J}|}{J_{\mathrm{c}}(T, B, \boldsymbol{r})} \right)^n \frac{\boldsymbol{J}}{|\boldsymbol{J}|}$$

The power law represents the macroscopic transition from flux creep to flux flow. In the limit as n approaches infinity, it reduces to the Bean critical state; for $n = 1$ the relation is ohmic (Rhyner, 1993). The criterion field is $E_{\mathrm{c}} = 1\ \mu\mathrm{V\ cm}^{-1}$. The 77 K self-field n value is selected from the range reported for commercial REBCO tape (Robert, Fareed and Ruiz, 2019).

Temperature, local magnetic field and position enter the critical current density through separate factors:

$$J_{\mathrm{c}}(T, B, \boldsymbol{r}) = k_{\mathrm{w}}(\boldsymbol{r})\, J_{\mathrm{c}0}\, f_T(T)\, f_B(B)$$

The temperature dependence is represented by the normalised power law

$$f_T(T) = \begin{cases} \left(\dfrac{T_\mathrm{c} - T}{T_\mathrm{c} - T_\mathrm{b}}\right)^\alpha, & T_\mathrm{b} \le T < T_\mathrm{c} \\ 0, & T \ge T_\mathrm{c} \end{cases}$$

Magnetic-field dependence is described by the anisotropic Kim relation (Robert, Fareed and Ruiz, 2019; Kim, Hempstead and Strnad, 1962; Zermeño et al., 2017; Grilli et al., 2014). It extends the inverse field dependence introduced by Kim, Hempstead and Strnad (1962):

$$f_B(B) = \frac{B_0}{B_0 + \sqrt{k_\mathrm{an}^2 B_\parallel^2 + B_\perp^2}}$$

The weak zone is introduced through a window function along the tape:

$$k_\mathrm{w}(x) = 1 - d[\Theta(x - x_\mathrm{c} + w_x) - \Theta(x - x_\mathrm{c} - w_x)], \qquad d = 0.15$$

Within this section, J_c is 85% of its base value over the tape section centred at $x_\mathrm{c} = 25$ mm with half-width $w_x = 2.5$ mm, and $k_\mathrm{w} = 1$ elsewhere.

E_c is the electric-field criterion for critical current, while n controls how E varies with J. $J_{\mathrm{c}0}$ denotes the reference critical current density at bath temperature and self-field. The exponent α determines how rapidly J_c falls with temperature, B_0 defines the field scale, and k_an describes the anisotropy between $B_\parallel$ and $B_\perp$. Layered REBCO is more sensitive to $B_\perp$. For SCS4050, Robert, Fareed and Ruiz (2019) fit $B_0 = 42.65$ mT, $k_\mathrm{an} = 0.295$ and $\alpha = 0.7$. Their temperature exponent differs from that used here. Field-temperature scaling measurements of commercial coated conductors are reported by Senatore et al. (2016).

The weak-zone parameters (d, x_c, w_x) represent longitudinal non-uniformity as a deterministic region of reduced critical current. This region provides a defined quench-initiation site and avoids the simultaneous transition produced by an ideally uniform tape. Colangelo and Dutoit (2012) examine the effect of tape non-uniformity on resistive SFCL behaviour.

The base case retains the prescribed weak-zone profile, whereas the control sets kw(x) = 1 throughout the tape. No other constitutive parameter is changed.

When the fault current drives J/J_c above unity, the power law increases both electric field and Joule heating. The associated temperature rise and local magnetic field reduce J_c, allowing the quench to extend beyond the weak zone. This electrothermal feedback determines the rapid resistance increase in the first half cycle and the slower growth thereafter.

Far above the critical current, a bounded transition first follows the flux-flow branch and then approaches the normal-state resistivity of the stack. This prevents an unbounded electric field during a severe fault while retaining the overcritical response.

A regularisation floor is applied to the subcritical resistivity solely for numerical conditioning; it is excluded from the device-resistance results.

3.2.3 Heat conduction

Transient heat conduction is solved only in the solid stack:

$$\rho_{\mathrm{m}} c_p(T)\,\frac{\partial T}{\partial t} = \nabla \cdot (k(T)\,\nabla T) + Q$$

$$Q = \boldsymbol{E} \cdot \boldsymbol{J}$$

Here, ρ_{m} denotes material density, $c_p(T)$ specific heat capacity at constant pressure and $k(T)$ thermal conductivity; each is temperature dependent. The source term Q is the Joule power density obtained from the electromagnetic solution. Heat transfer is not solved in the air domain because air conduction is negligible on the millisecond fault timescale.

3.2.4 Liquid-nitrogen boundary condition

Pool boiling at each liquid-nitrogen-wetted surface is represented by a heat-transfer coefficient that depends on wall superheat:

$$q = h_{\mathrm{LN_2}}(\Delta T)\,\Delta T$$

$$\Delta T = T_{\mathrm{wall}} - T_{\mathrm{bath}}$$

Here, q denotes wall heat flux, T_{wall} solid-wall temperature and T_{bath} liquid-nitrogen bath temperature. The coefficient $h_{\mathrm{LN_2}}$ is obtained from a steady pool-boiling curve.

3.2.5 External circuit model

In the physical connection shown in Figure 3.2, the upstream grid supplies a 400 V point of common coupling through a 50 kVA transformer. A 35 kW synchronous generator connects at the same bus. The SFCL and circuit breaker are placed in series with the downstream feeder, so the limiter carries the fault current.

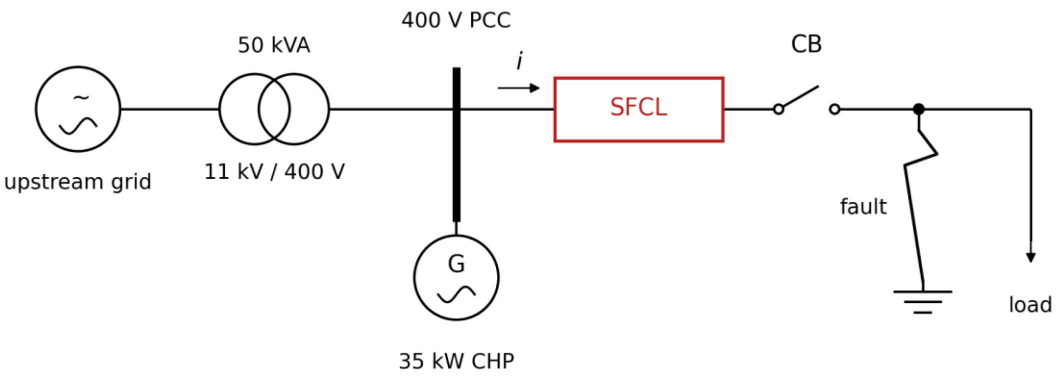

Figure 3.2 Position of the SFCL in the 400 V grid-connected feeder.

The circuit uses the single-phase Thévenin equivalent in Figure 3.3. The source parameters are $R_\mathrm{s} = 0.0314\ \Omega$ and $\omega L_\mathrm{s} = 0.1568\ \Omega$, giving $X/R = 3.79$. The feeder load is $R_\mathrm{load} = 4.503\ \Omega$. The three-dimensional field model supplies the SFCL terminal voltage to the circuit, while a time-dependent resistance represents fault application and breaker opening.

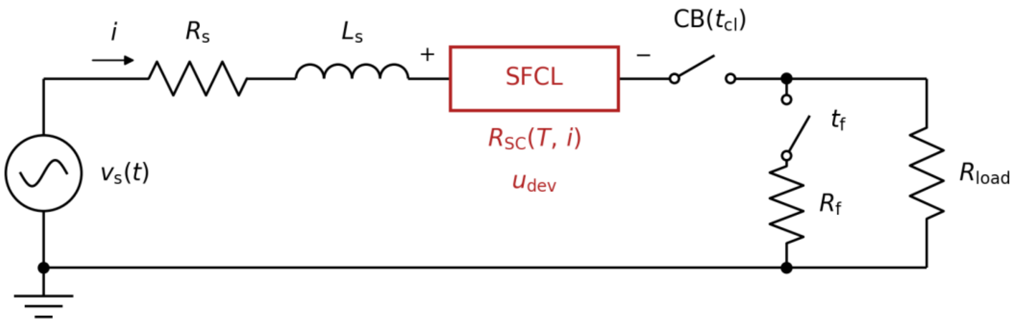

Figure 3.3 External-circuit equivalent used in the field-circuit computation.

The circuit is governed by the following Kirchhoff voltage equation and time-dependent load resistance:

$$v_\mathrm{s}(t) = R_\mathrm{s}\, i + L_\mathrm{s}\frac{\mathrm{d}i}{\mathrm{d}t} + u_\mathrm{dev}(i, T) + R_L(t)\, i$$

$$R_L(t) = R_\mathrm{load} - (R_\mathrm{load} - R_\mathrm{f})s(t;\ t_\mathrm{f}, \tau_\mathrm{s}) + R_\mathrm{open}\, s(t;\ t_\mathrm{cl}, \tau_\mathrm{s})$$

Here, $s(\cdot)$ is a twice-continuously differentiable smoothed step with transition half-width $\tau_\mathrm{s} = 1$ ms. The resistance values are $R_\mathrm{load} = 4.503\ \Omega$, $R_\mathrm{f} = 0.010\ \Omega$ and $R_\mathrm{open} = 450.3\ \Omega$. The fault is initiated at $t_\mathrm{f} = 10$ ms. All fault quantities use this inception phase, and the clearing instant t_cl is varied parametrically.

The two parallel conductor strings satisfy current continuity and equal terminal voltage. Each string contains two tape runs in series, while the field model resolves the four runs in positive, negative, positive and negative order through the stack. A length factor of 1500, derived from the device geometry, maps the 50 mm segment to the complete limiter. The circuit, magnetic-field and heat-transfer equations are solved within the same transient step, avoiding a lag between sub-models.

Continuous step functions introduce the fault and breaker resistance without the oscillation associated with discontinuous switching. The open breaker is represented by a large finite resistance, leaving a negligible residual current in the numerical solution.

3.3 Geometry and material stack

The limiting element uses double-sided copper-stabilised REBCO tape, 4 mm wide and 95 µm thick. From one face to the other, the stack comprises 20 µm copper, 2 µm silver, 50 µm Hastelloy, 1 µm REBCO, 2 µm silver and 20 µm copper. Four tapes form the representative segment. Adjacent runs carry opposing currents, which reduces net loop flux linkage and stored magnetic energy while retaining the local near field and associated AC loss. The geometry is shown in Figure 3.4.

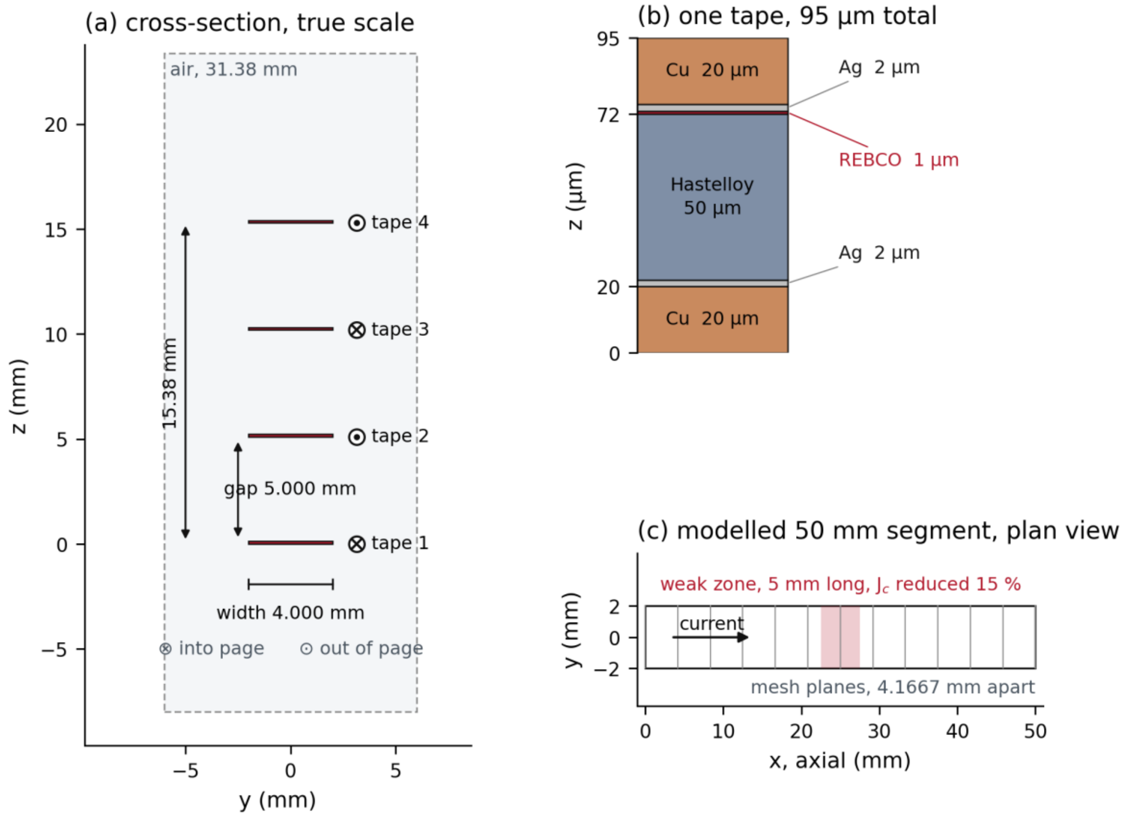

Figure 3.4 Tape stack and modelled segment geometry.

The number of parallel tapes follows from the rated-current requirement and the critical-current margin. The model assigns a self-field critical current of 72 A at 77 K to each 4 mm tape. Under the same criterion, five commercial tapes from four manufacturers carry 116–226 A, including 116 A for SCS4050-CF and 156 A for SCS4050-AP (Tsuchiya et al., 2017). The model value is approximately 38% below the lowest measurement in that group. Design guidance limits rated current to 40% of critical current (Tixador, 2023), while the total device critical current must also exceed the peak associated with a 120% rated load (Colangelo and Dutoit, 2012).

For a rated current of 51 A rms, the steady-state margin requires a total critical current of $51/0.4 = 127.5$ A. The peak current at 120% load is $1.2 \times \sqrt{2} \times 51 \approx 86.5$ A, requiring at least 86.5 A. One 72 A tape is therefore insufficient; two tapes in parallel provide 144 A. The 51 A rms operating point is 35.4% of this DC critical current and satisfies both criteria.

Each parallel branch is an outgoing and return series counter-winding. Opposing current directions reduce the branch flux linkage and inductance, but retain the local field between adjacent tapes. Two such branches give four current-carrying runs in the 50 mm segment; Figure 3.5 gives their three-dimensional arrangement and current directions.

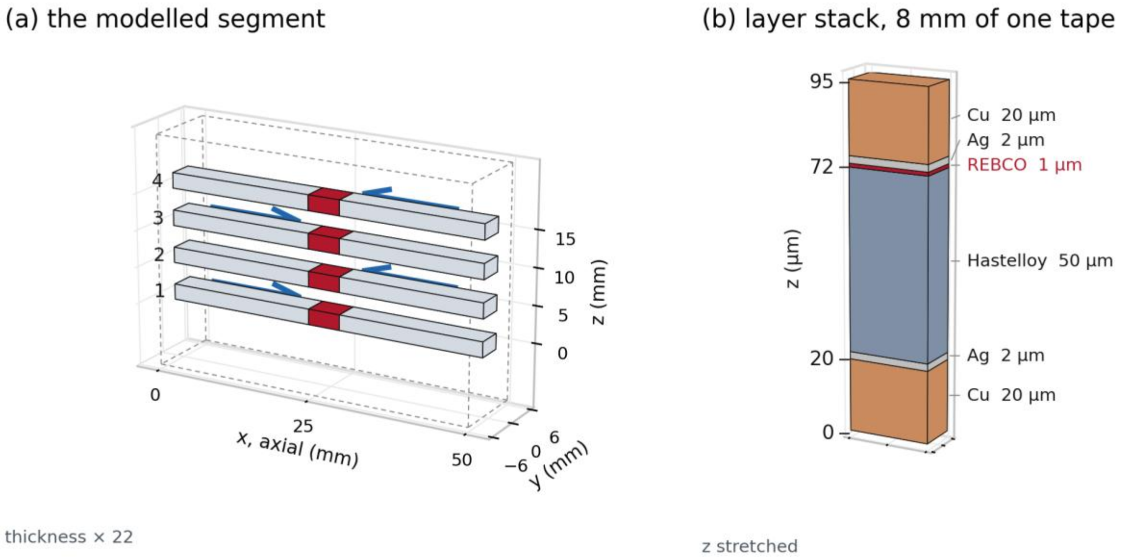

Figure 3.5 Three-dimensional layout of the counter-wound stack and single-tape layer order.

All derived dimensions depend parametrically on the inter-tape gap. The principal values are listed in Table 3.1.

Table 3.1 Core geometric parameters.

Quantity	Symbol	Value
Single-tape total thickness	t_{total}	0.095 mm
Inter-tape gap	gap_z	5 mm
Inter-tape pitch	$t_{\mathrm{total}} + \mathrm{gap}_z$	5.095 mm
Four-tape stack height	N/A	15.38 mm
z-direction air-domain margin	air_margin_z	8 mm
Air-domain height	air_Lz	31.38 mm
Modelled segment length	N/A	50 mm
Tape width	N/A	4 mm

3.4 Discretisation and solution

The model is implemented in COMSOL Multiphysics 6.4.0.293. The magnetic-field formulation uses $\boldsymbol{H}$ as its dependent variable in the mfh interface. Heat transfer in the solids is solved with the ht interface and is coupled bidirectionally through the volumetric Joule source. Lumped circuit elements are advanced within the same continuous transient solution; no isothermal approximation or segmented time window is used.

The spatial discretisation uses finite elements and the transient solution uses backward differentiation. Table 3.2 lists the principal solver settings.

Table 3.2 Core solver settings.

Setting	Value
Mesh	Production mesh (12 axial elements, 13 axial planes)
Elements	119,184
Degrees of freedom	554,999
Relative tolerance	1×10^{-4}
Time stepping	BDF, maximum order 2
Output step	2×10^{-4} s
Solution window	Single continuous transient solve, no time-axis segmentation
Liquid-nitrogen bath temperature	77 K
Open-breaker equivalent resistance	450.3 Ω

The mesh is generated by sweeping a two-dimensional cross-section through 12 axial layers. The cross-section contains 8,972 triangular and 960 quadrilateral elements, giving 119,184 elements, 71,201 nodes and 554,999 degrees of freedom in three dimensions. A single element spans the 1 μm superconducting layer. Axial refinement does not alter the transverse or through-thickness discretisation. Figure 3.6 shows the production mesh.

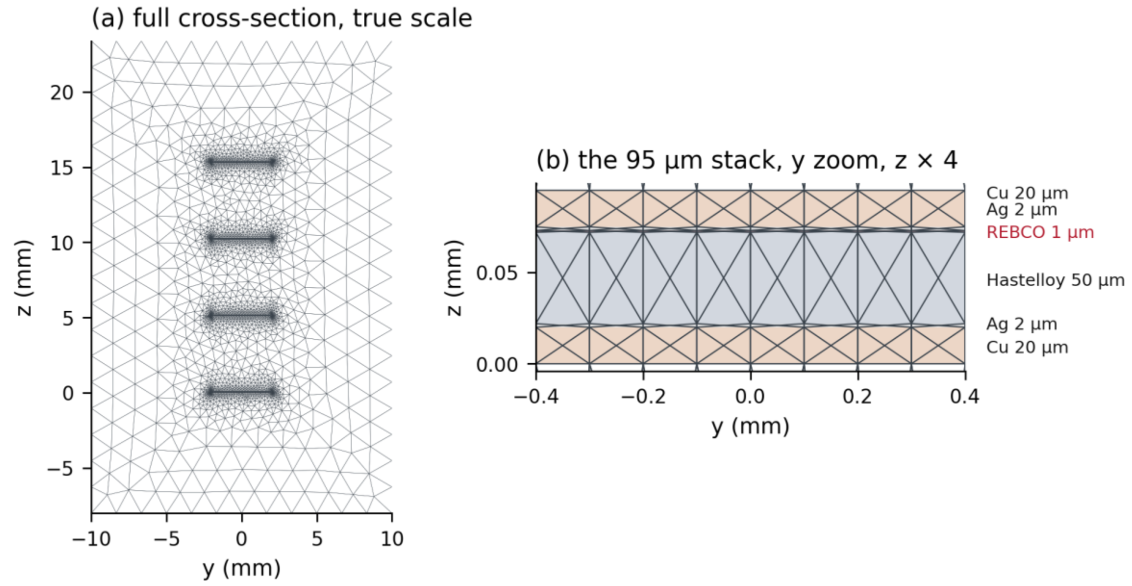

Figure 3.6 Production-mesh cross-section with a close-up of the stack.

The maximum coupling operator, maxop, extracts the superconducting-layer hot-spot temperature. A volume-maximum operator, MaxVolume, provides a sampling-sensitivity check over the same domain with a different set of integration points. Their comparison is reported in §4.4.

The two cases use the same geometry, mesh, external circuit, initial temperature, fault timing, solver and output instants; only kw(x) differs.

3.5 AC loss computation

Rated-current AC loss is calculated in a separate, fault-free periodic case using the same voltage source, geometry and mesh. The breaker remains closed for several power-frequency cycles. The external circuit determines the line current, with 51 A rms used as the design operating point, and the loss is integrated over the final complete cycle.

The transport AC loss per unit length, per tape and per cycle reads

$$Q_{\mathrm{AC}} = \frac{1}{N_{\mathrm{tape}} L} \int_{t_0}^{t_0+T} \int_V \boldsymbol{E} \cdot \boldsymbol{J} \, \mathrm{d}V \, \mathrm{d}t$$

Here, T denotes the power-frequency period, $L = 0.05$ m is the segment length, $N_{\mathrm{tape}} = 4$ is the number of current-carrying runs, and t_0 is the start of the integration interval. The domain V covers all solid layers of the four runs, and the result is normalised by the 0.2 m total tape length.

4. Results and Analysis

4.1 Base case and data conventions

The 50 ms base case is solved continuously over 0–200 ms with results stored every 0.2 ms. Fault insertion occurs over 9.0–11.0 ms and breaker opening over 59.0–61.0 ms. The line current, terminal voltage, temperature and equivalent resistance are taken from this single coupled solution.

Results are reported in terms of line current I, device terminal voltage U and parallel equivalent resistance $R = U/I$. Each parallel string carries $I/2$, and its resistance is twice the parallel value.

4.2 Line current

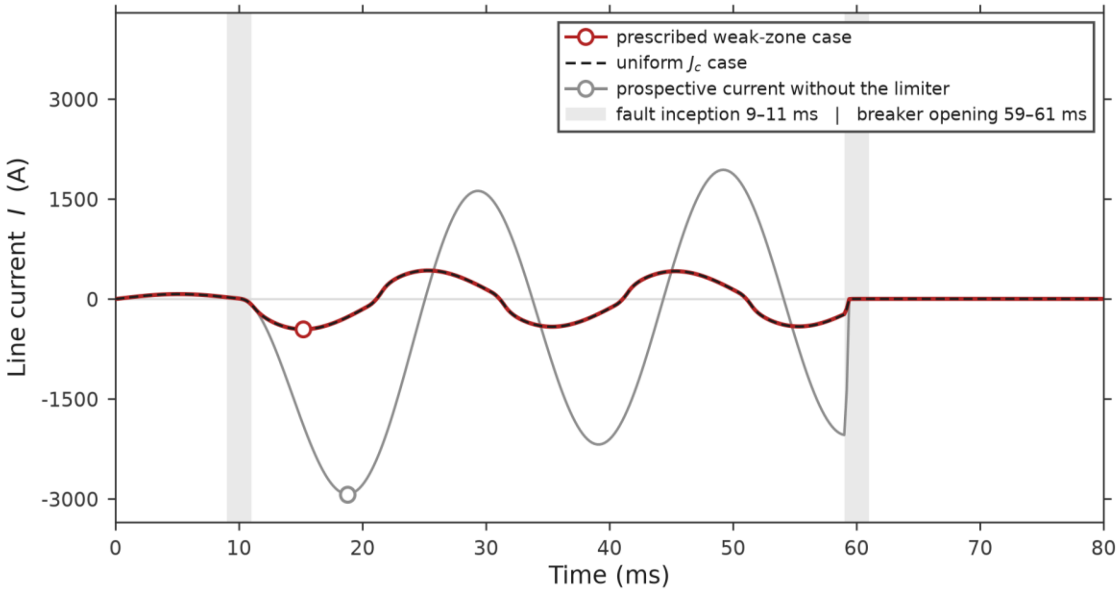

Figure 4.1 Line current for the weak-zone and uniform-Jc SFCL cases, with the prospective current shown for reference.

As shown in Figure 4.1, the pre-fault current peaks at approximately 72 A. Fault application raises the current to 456.1 A at 15.2 ms during the first half cycle, after which the envelope declines gradually to about 413 A before breaker opening. The non-sinusoidal terminal voltage produces the kinks around the current zero crossings.

The zero-impedance reference reaches a first peak of 2.93 kA; the SFCL reduces this by 84.4%. The reference and limited peaks are separated by 3.6 ms because the loop impedance angle changes as X/R decreases from 3.79 to approximately 0.24. Varying the fault-inception phase changes the reduction by about 4.3 percentage points, making the phase convention necessary when comparing first peaks.

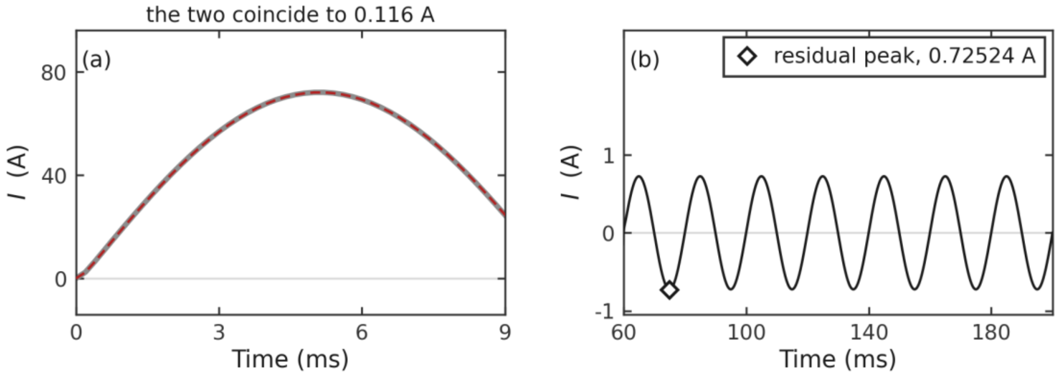

Figure 4.2 Low-amplitude current: (a) normal operation; (b) after breaker opening.

Figure 4.2 enlarges the low-current intervals that are unresolved in the main plot. The normal-operation currents before and after SFCL insertion are nearly identical. Once the breaker opens, the finite open-state resistance leaves a residual current below 1 A.

4.3 Device terminal voltage

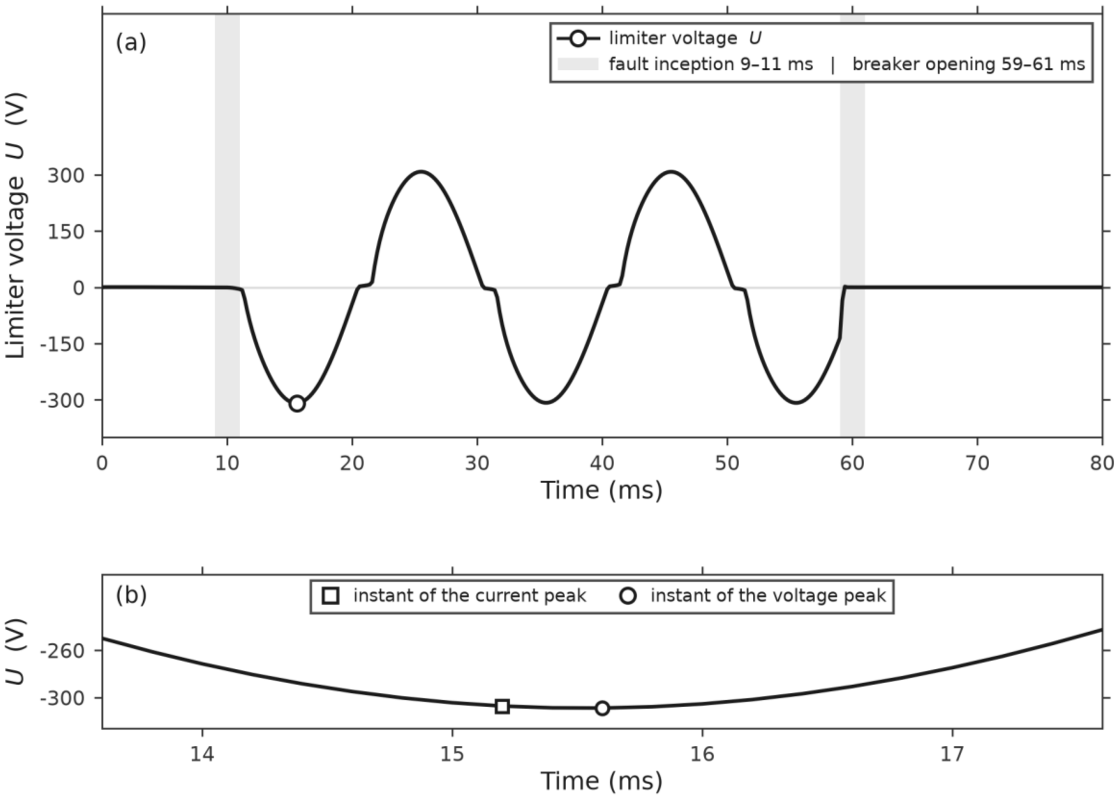

Figure 4.3 SFCL terminal voltage with a close-up of the first peak.

The terminal voltage reaches 308.9 V at 15.6 ms, 0.4 ms after the current peak. The enlargement in Figure 4.3 confirms that both extrema occur within the first fault half cycle but not at the same instant.

The terminal voltage approaches zero at each current zero and rises again with the transport current. During later half cycles, increasing resistance is offset by the declining current envelope, so the voltage remains below its first peak. Dividing the terminal voltage by the 150 m effective length gives the device-average electric field.

4.4 Temperature and local electrothermal distribution

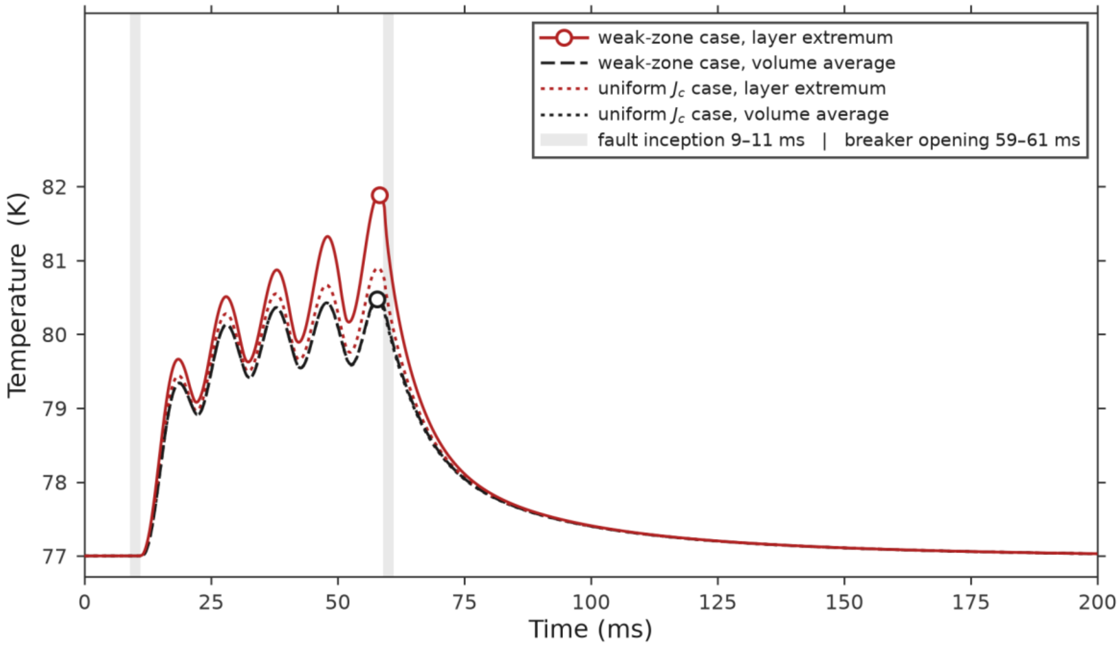

Figure 4.4 Volume-average and superconducting-layer extremum temperatures for the weak-zone and uniform-Jc cases.

The volume-average temperature reaches 80.48 K at 57.8 ms. The superconducting-layer hot spot follows 0.4 ms later at 81.89 K, as shown in Figure 4.4. Their separation increases during the fault until breaker opening terminates further heating.

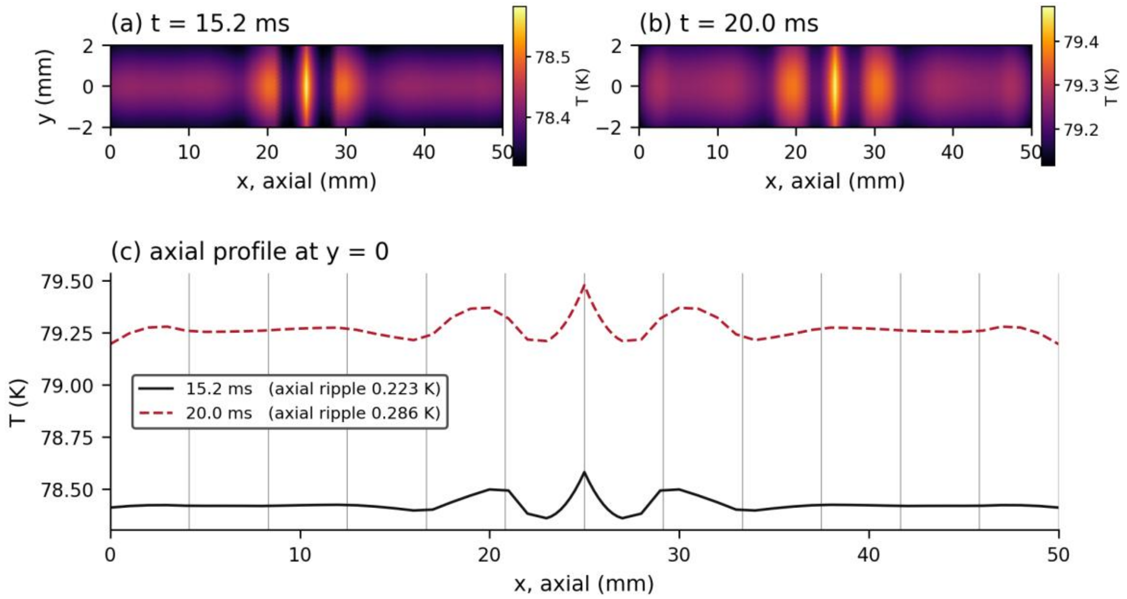

Figure 4.5 Temperature over the superconducting plane (50 ms clearing case).

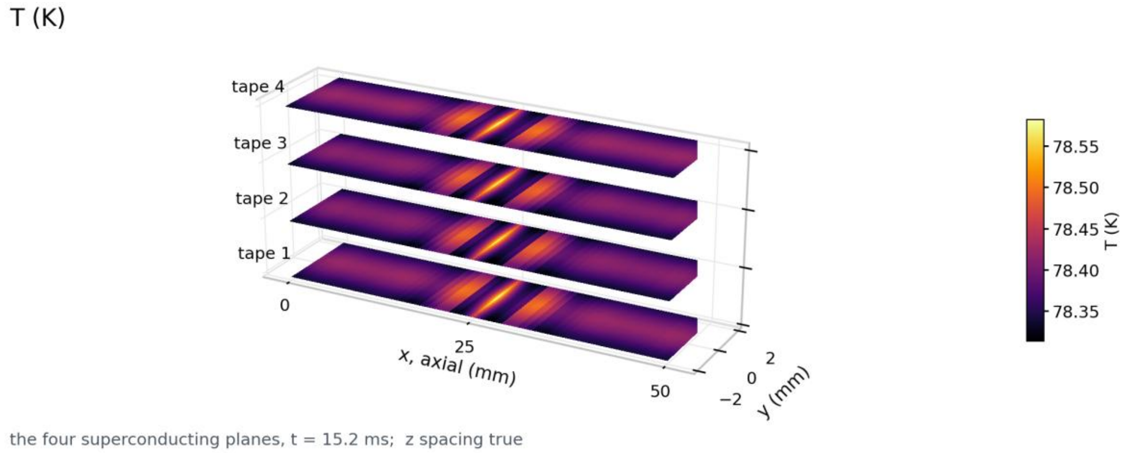

Figure 4.6 Temperature over the four superconducting planes (15.2 ms, 50 ms clearing case).

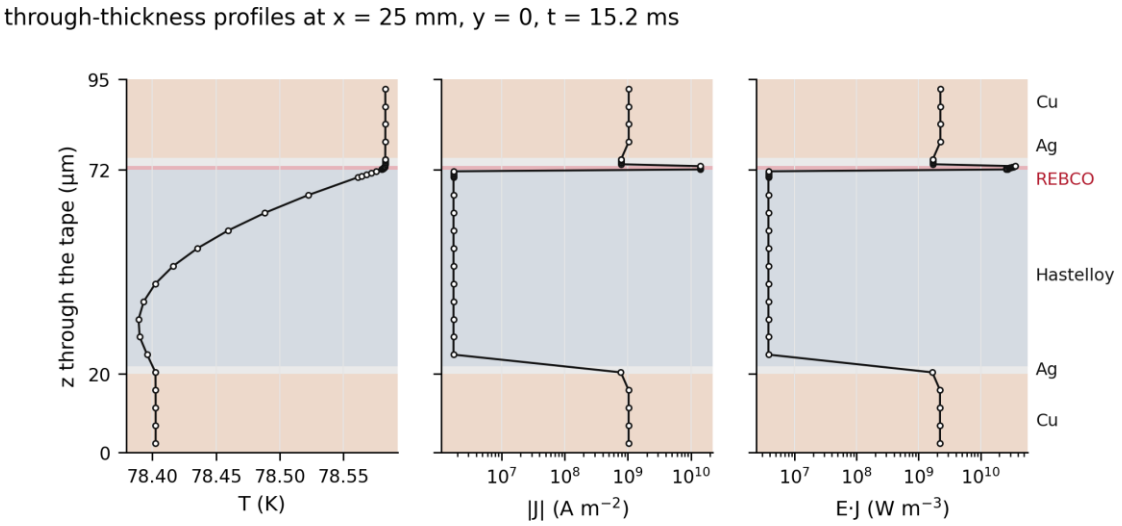

Figure 4.7 Through-thickness temperature, current density and volumetric power density (x = 25 mm, 15.2 ms).

The spatial distributions in Figures 4.5 and 4.6 locate the hot spot in the superconducting layer. Through the thickness, Figure 4.7 shows current shared mainly by the REBCO and copper stabiliser, whereas Joule dissipation is concentrated in the REBCO. The surface-average heat flux, used as the device-level heat-transfer measure, reaches approximately 3.15×10^4 W/m² in the base case.

4.5 Equivalent resistance and current density

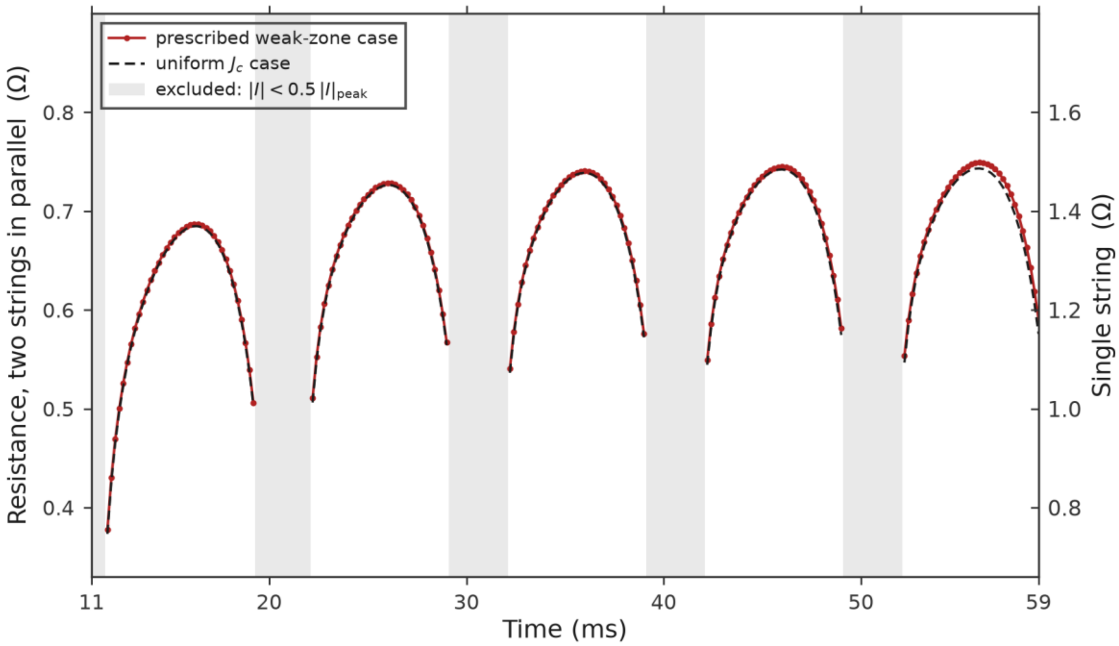

Figure 4.8 Device equivalent resistance for the weak-zone and uniform-Jc cases.

Figure 4.8 presents U/I after full fault application and before breaker operation; samples around the current zero crossings are excluded. The parallel equivalent resistance is approximately 0.674 Ω near the first current peak and increases gradually to 0.745 Ω before opening. The accompanying decline in the current envelope reflects the temperature dependence of device impedance.

At a current extremum, $\mathrm{d}I/\mathrm{d}t = 0$, so the inductive contribution to U/I is minimised. Figures 4.9 and 4.10 locate current-density concentration at the tape edges and around the weak zone. The local ratio J/J_c marks quench initiation, whereas current margin is assessed from the single-tape transport current relative to critical current.

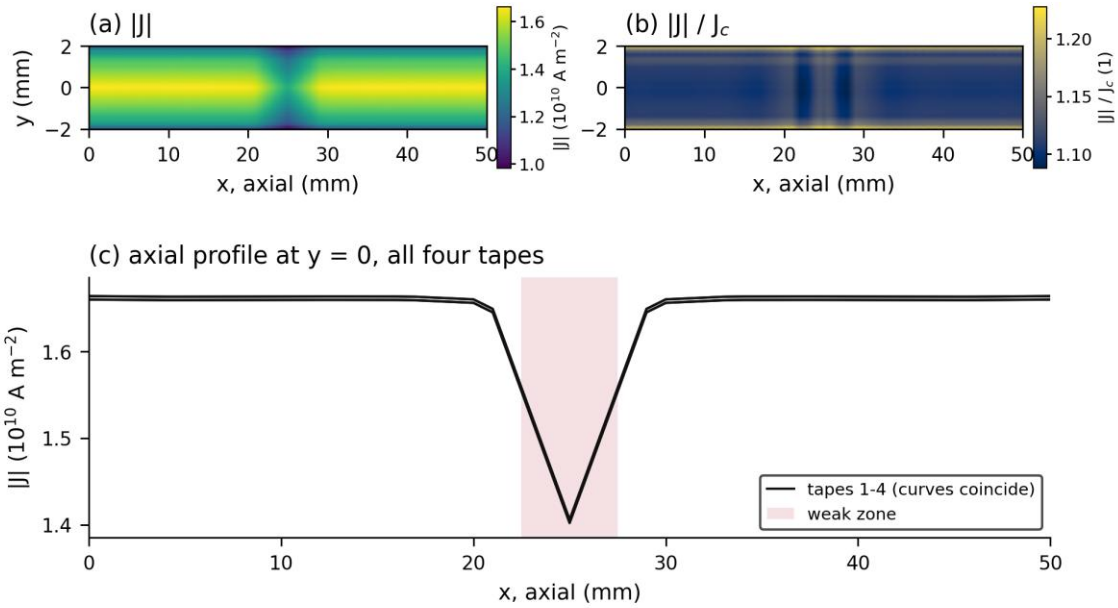

Figure 4.9 Current density over the superconducting plane (15.2 ms, 50 ms clearing case).

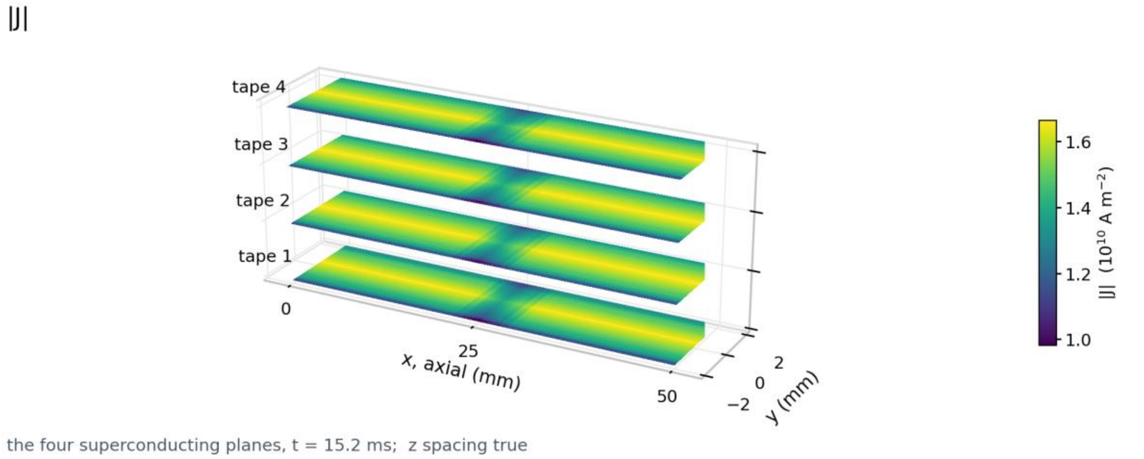

Figure 4.10 Current density over the four superconducting planes (15.2 ms, 50 ms clearing case).

4.6 Effect of the prescribed weak zone

The comparison isolates the prescribed weak-zone pattern by setting kw(x) = 1 in the control case while retaining the common settings defined in §3.4. Because the 50 mm cell is length-scaled to the complete device, the base model represents a periodic weak-zone pattern rather than an isolated defect. Table 4.1 shows that the uniform-Jc control raises the first current peak by 0.31%, while the terminal-voltage peak is unchanged at the reported precision.

Table 4.1 Electrical and thermal comparison of the weak-zone and uniform-Jc cases.

Quantity	Prescribed weak zone	Uniform Jc	Change
First fault-current peak	456.1 A (15.2 ms)	457.5 A (15.2 ms)	+1.4 A (+0.31%)
Terminal-voltage peak	308.9 V (15.6 ms)	308.9 V (15.6 ms)	No change at reported precision
Superconducting-layer extremum	81.89 K (58.2 ms)	80.90 K (58.0 ms)	−0.99 K
Volume-average peak	80.48 K (57.8 ms)	80.45 K (57.8 ms)	−0.03 K
Equivalent resistance near first peak	0.674 Ω (15.2 ms)	0.671 Ω (15.2 ms)	−0.31%
Cumulative Joule heat to interruption	2.23 J (61.0 ms)	2.22 J (61.0 ms)	−0.41%
Window-end electrical residual	−0.81%	−0.070%	—
Axial temperature distribution through 20 ms	Maximum at prescribed x = 25.0 mm zone	No prescribed site; axial spread ≤0.08 K	—

Equivalent resistance near the first current peak and cumulative Joule heat to interruption fall by 0.31% and 0.41%, respectively. The window-end electrical residuals are −0.81% for the weak-zone case and −0.070% for the uniform-Jc case. At device level, the two responses remain close for this operating condition.

The larger difference occurs in the local thermal response. The superconducting-layer extrema are 81.89 K and 80.90 K for the weak-zone and uniform-Jc cases, respectively, and their 0.99 K separation exceeds the 0.6 K cross-case sampling resolution. The corresponding volume-average peaks remain close at 80.48 K and 80.45 K. At 15.2 ms, Figure 4.11 shows a zone-centred temperature band in the weak-zone case, whereas the uniform-Jc plane remains nearly uniform in the axial direction. The local J/Jc depression in the same region is consistent with current sharing into the stabiliser.

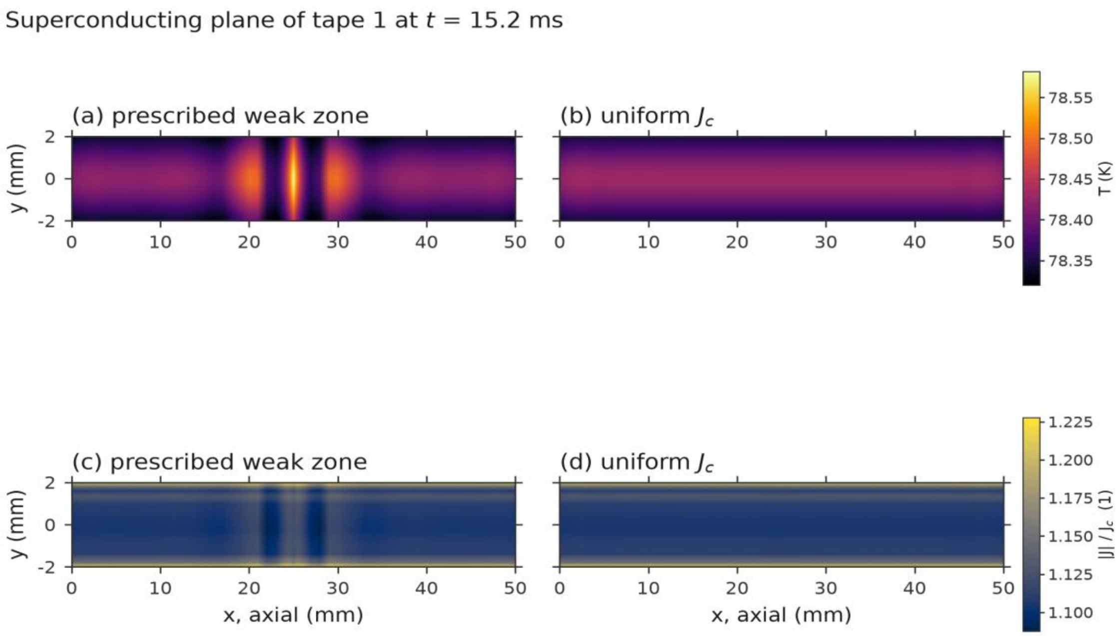

Figure 4.11 Temperature and J/Jc over the superconducting plane of tape 1 at 15.2 ms: (a, c) weak-zone case; (b, d) uniform-Jc case.

Figure 4.12 follows the axial temperature profile through the first 20 ms of the fault. A temperature maximum is centred on the prescribed zone at 12.0, 15.2 and 20.0 ms, while the uniform-Jc profile remains nearly flat. Mesh-related axial oscillations account for 14.1% and 11.5% of the weak-zone profile variation at 15.2 and 20.0 ms, respectively; the zone-centred temperature concentration remains the robust spatial feature.

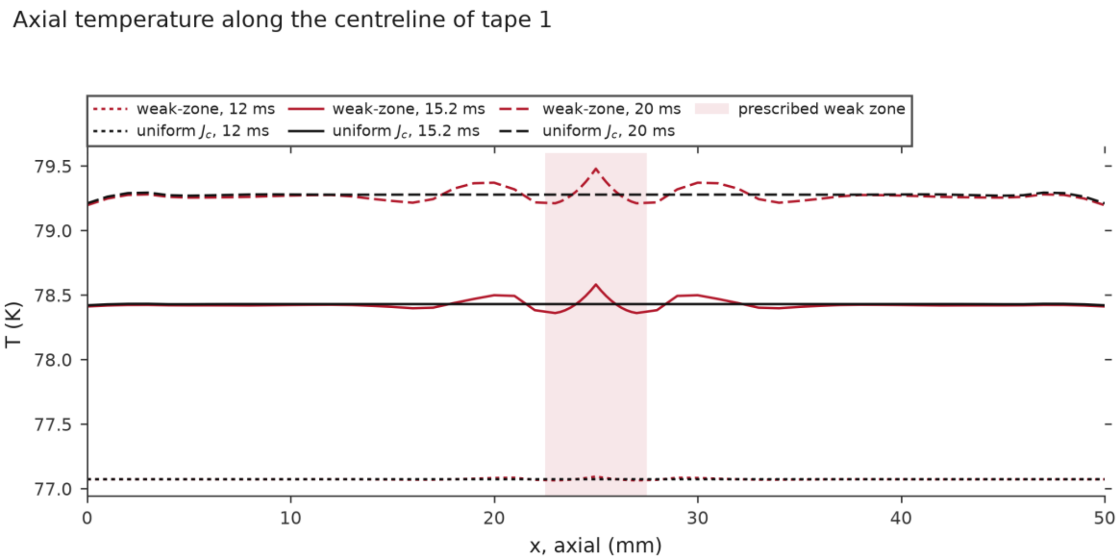

Figure 4.12 Axial temperature along the centreline of tape 1 at 12.0, 15.2 and 20.0 ms.

4.7 Energy balance

The base-case energy terms are evaluated from the following integrals. Here, $t_{\mathrm{w}} = 200$ ms, Ω_{s} denotes the solid domain and $\partial\Omega_{\mathrm{c}}$ the cooled surface:

$$W_{\mathrm{in}} = \int_0^{t_{\mathrm{w}}} u_{\mathrm{dev}}\ i\ \mathrm{d}t$$

$$Q_{\mathrm{J}} = \int_0^{t_{\mathrm{w}}} \int_{\Omega_{\mathrm{s}}} \boldsymbol{E} \cdot \boldsymbol{J}\ \mathrm{d}V\,\mathrm{d}t$$

$$\Delta H = \int_{\Omega_{\mathrm{s}}} \rho_{\mathrm{m}}\, c_p (T - T_{\mathrm{b}}) \mathrm{d}V \bigg|_{t_{\mathrm{w}}}$$

$$Q_{\mathrm{out}} = \int_0^{t_{\mathrm{w}}} \int_{\partial\Omega_{\mathrm{c}}} q\ \ \mathrm{d}A\,\mathrm{d}t$$

$$\Delta W_{\mathrm{mag}} = \frac{1}{2} \int_{\Omega} \boldsymbol{B} \cdot \boldsymbol{H}\ \mathrm{d}V \bigg|_0^{t_{\mathrm{w}}}$$

Figure 4.13 records approximately 2.23 J of cumulative Joule dissipation. By the end of the window, most of this energy has been transferred to the liquid nitrogen and only 4.45×10^{-3} J remains as solid enthalpy rise. The difference between Joule heat and the sum of enthalpy rise and surface heat removal is 0.27%.

Electrical-port energy is 0.81% lower than the volumetric Joule heat. The stored magnetic energy is much smaller than this difference, and changing the time-integration scheme changes the residual only to 0.80%. The remaining discrepancy is therefore treated as numerical error.

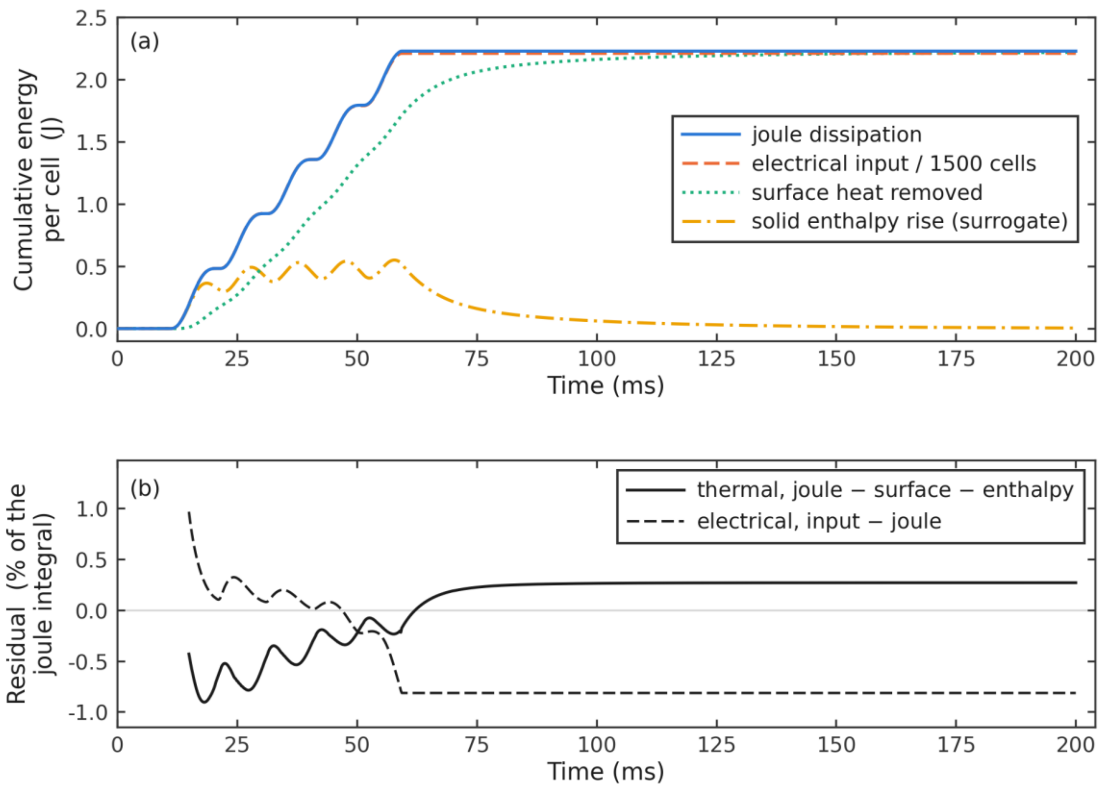

Figure 4.13 Energy balance of the base case.

4.8 Influence of the fault clearing time

Table 4.2 compares seven clearing times. Interruption at 2 or 5 ms occurs before the fault current is fully established and produces little heating. Between 20 and 60 ms, both the volume-average and hot-spot temperatures rise steadily with fault exposure. At 80 ms the volume-average temperature reaches 82.18 K and the boundary condition enters its prescribed boiling-transition regime, so this case follows a different heat-transfer branch.

Table 4.2 Peak temperatures of the seven clearing cases.

Fault clearing time	Extremum temperature: maximum coupling operator	Extremum temperature: volume-maximum operator, refined-sampling converged value	Volume-average temperature
2 ms	77.00 K (11.2 ms)	77.00 K (11.2 ms)	77.00 K (11.4 ms)
5 ms	77.97 K (14.2 ms)	77.97 K (14.2 ms)	77.77 K (14.2 ms)
20 ms	80.51 K (28.0 ms)	80.51 K (28.0 ms)	80.13 K (28.0 ms)
40 ms	81.33 K (48.0 ms)	81.33 K (48.0 ms)	80.43 K (47.8 ms)
50 ms	81.89 K (58.2 ms)	81.89 K (58.2 ms)	80.48 K (57.8 ms)
60 ms	83.14 K (68.8 ms)	83.14 K (68.8 ms)	80.57 K (68.0 ms)
80 ms	heat-transfer transition regime	heat-transfer transition regime	82.18 K (88.0 ms)

The electrical extrema in Table 4.3 show that clearing times of 20 ms and above do not change the first current peak; they govern only the subsequent energy deposition. The 80 ms case is the sole exception in terminal voltage, reaching 312.0 V at 85.6 ms late in the fault. With 2 and 5 ms clearing, interruption is complete before the principal peak develops.

Table 4.3 Electrical extrema of the seven clearing cases.

Fault clearing time	Current interruption instant	Window maximum line current	Window maximum terminal voltage
2 ms	12 ms	88.4 A (11.2 ms)	9.4 V (11.2 ms)
5 ms	15 ms	430.9 A (14.0 ms)	271.5 V (14.0 ms)
20 ms	30 ms	456.1 A (15.2 ms)	308.9 V (15.6 ms)
40 ms	50 ms	456.1 A (15.2 ms)	308.9 V (15.6 ms)
50 ms	60 ms	456.1 A (15.2 ms)	308.9 V (15.6 ms)
60 ms	70 ms	456.1 A (15.2 ms)	308.9 V (15.6 ms)
80 ms	90 ms	456.1 A (15.2 ms)	312.0 V (85.6 ms)

The $\int I^2 \mathrm{d}t$ values in Table 4.4 increase monotonically with clearing time. This integral reflects the duration of the established fault current and accounts for the ordering of the temperature rises.

Table 4.4 $\int I^2 \mathrm{d}t$ of the seven clearing cases.

Fault clearing time	$\int I^2 \mathrm{d}t$ from fault inception
2 ms	3.77 $A^2 \cdot s$
5 ms	344 $A^2 \cdot s$
20 ms	2.13×10^3 $A^2 \cdot s$
40 ms	4.02×10^3 $A^2 \cdot s$
50 ms	4.96×10^3 $A^2 \cdot s$
60 ms	5.87×10^3 $A^2 \cdot s$
80 ms	7.47×10^3 $A^2 \cdot s$

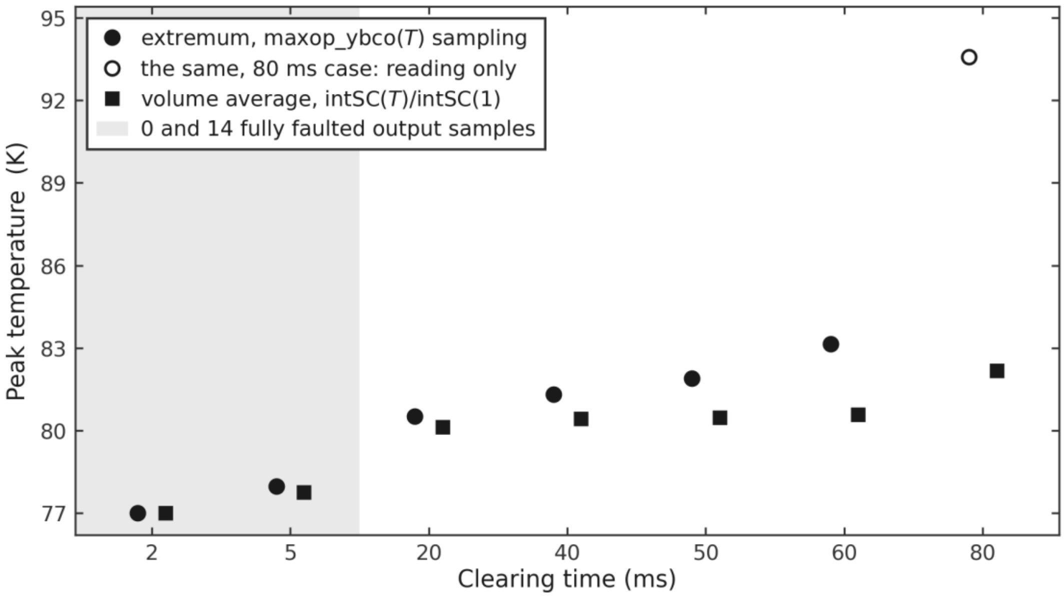

Figure 4.14 Volume-average and extremum temperature across clearing times.

4.9 Cooling after fault clearance

The threshold data in Table 4.5 show that most cases cool below 78 K within approximately 15–18 ms after interruption. The 80 ms case begins from the highest temperature and requires 25.0 ms.

Crossing 80 K belongs to the initial cooling stage and occurs within a few milliseconds for all cases, although the 80 ms case requires 11.2 ms. As the conductor approaches the bath temperature, the cooling rate decreases and the differences caused by clearing time diminish.

Table 4.5 Threshold crossing times after interruption.

Fault clearing time	Current interruption instant	78 K crossing (relative to interruption)	80 K crossing (relative to interruption)
20 ms	30 ms	45.2 ms (15.2 ms)	N/A
40 ms	50 ms	66.4 ms (16.4 ms)	52.0 ms (2.0 ms)
50 ms	60 ms	77.0 ms (17.0 ms)	62.8 ms (2.8 ms)
60 ms	70 ms	88.0 ms (18.0 ms)	74.2 ms (4.2 ms)
80 ms	90 ms	115.0 ms (25.0 ms)	101.2 ms (11.2 ms)

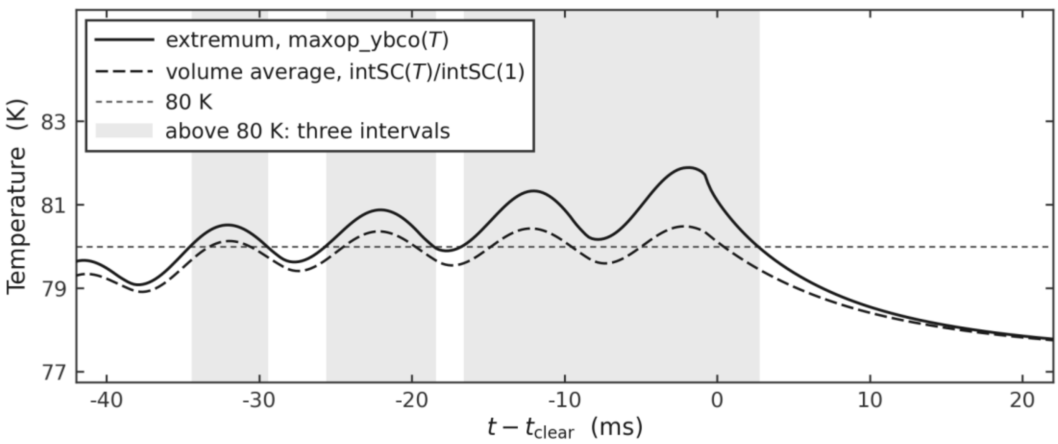

Figure 4.15 Temperature evolution referred to the interruption instant (50 ms clearing case).

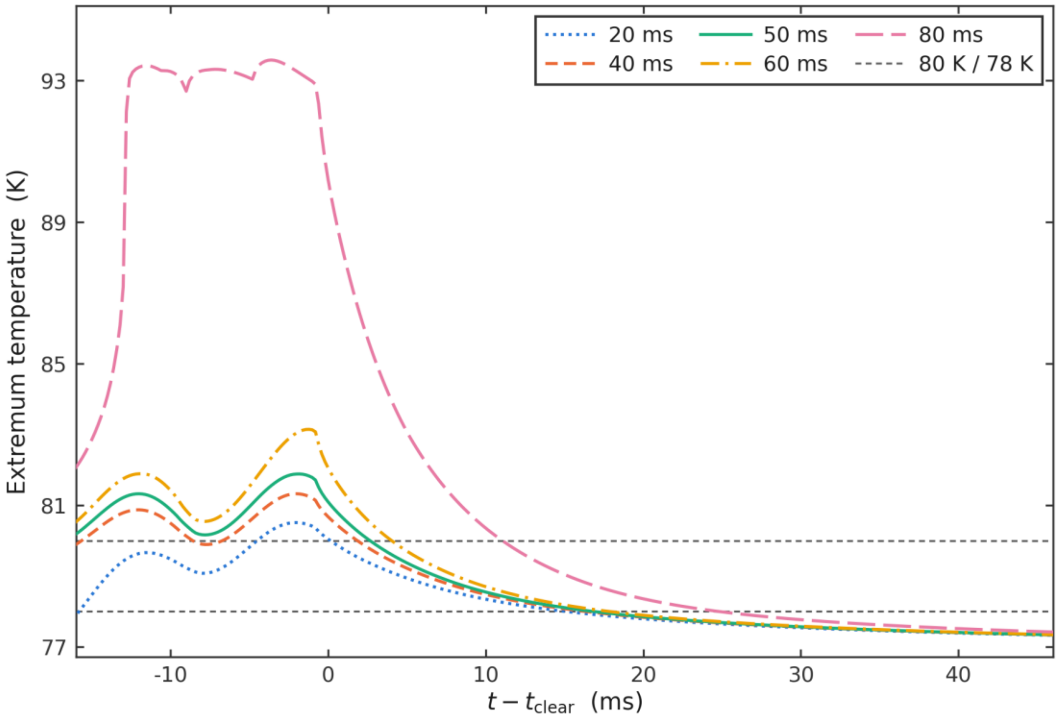

Figure 4.16 Extremum temperature of five clearing cases, each referred to its own interruption instant.

4.10 AC loss at rated current

The fault-free calculation covers six power-frequency cycles, with loss evaluated over the last complete cycle. The final-cycle line current is 50.9 A rms, close to the 51 A design point, and each parallel string carries approximately 25.4 A. The hot-spot temperature rises by less than 0.4 mK, so the conductor remains at essentially the same thermal state.

The loss is 3.51×10^{-4} J/(cycle·m) per tape, of which the superconducting layer contributes approximately 99.8%. Extrapolation to the 300 m device gives 5.27 W, compared with 5.26

W from the port-power integral; the 0.17% difference indicates the consistency of the two calculations. Song et al. (2018) also reported increased transport loss from a central gap in a bifilar stack, although for a different geometry and operating point.

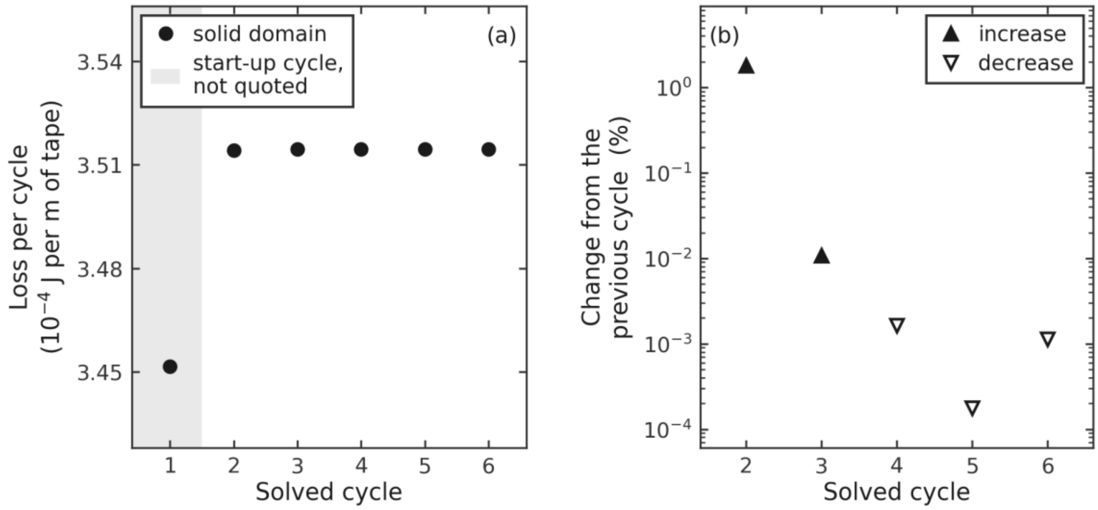

Figure 4.17 Cycle-by-cycle AC loss and inter-cycle drift at rated operation.

4.11 Magnetic field redistribution and loop inductance

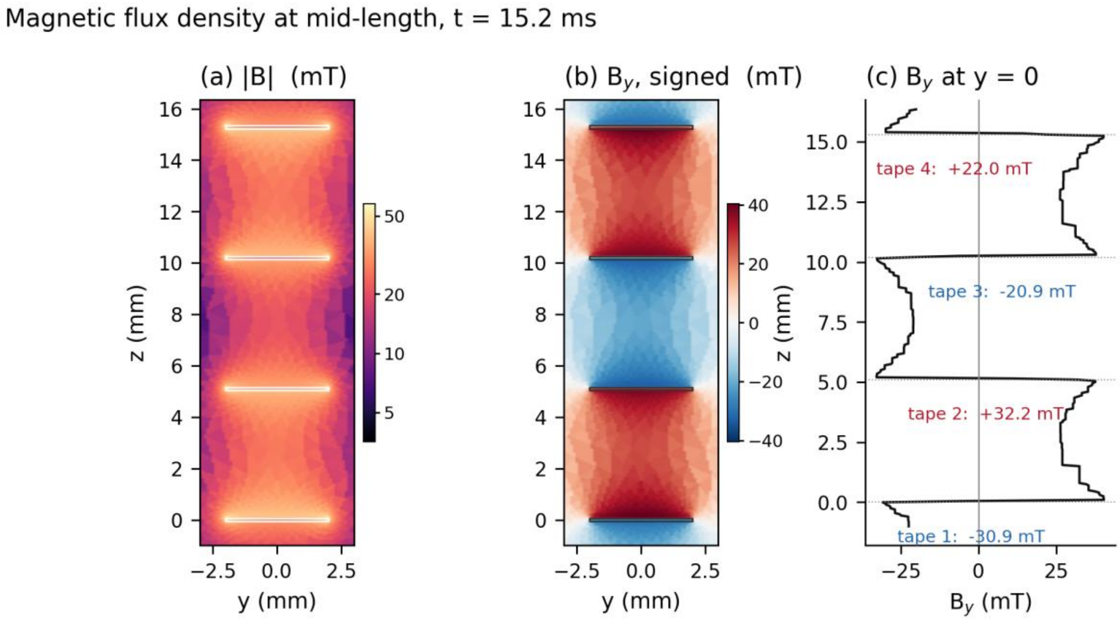

Figure 4.18 Cross-section flux density at the fault peak (15.2 ms, 50 ms clearing case).

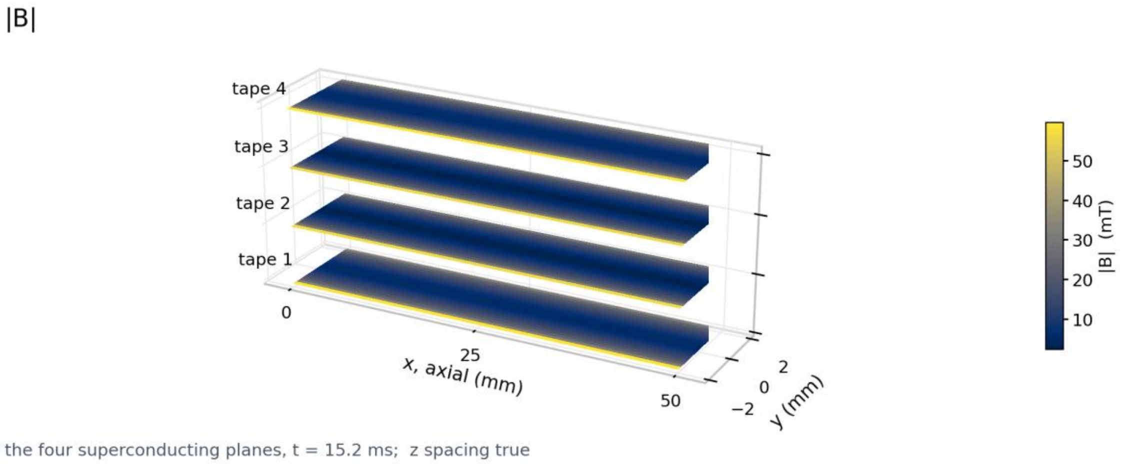

Figure 4.19 Flux density magnitude over the four superconducting planes (15.2 ms, 50 ms clearing case).

At the first fault peak, Figures 4.18 and 4.19 show a surface flux-density magnitude of approximately 57.7 mT. The axial component changes sign with the winding direction of the four runs, while 20–26 mT remains between neighbouring tapes. Counter-winding therefore reduces net loop flux linkage without eliminating the near field at the conductor.

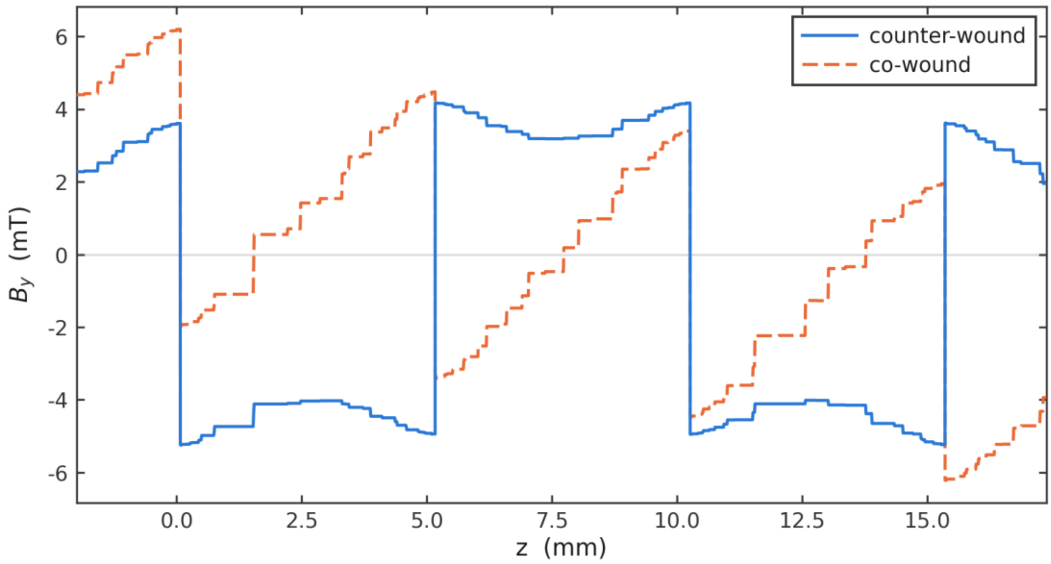

Figure 4.20 B_y through the four-tape stack, counter-wound against co-wound (rated operation).

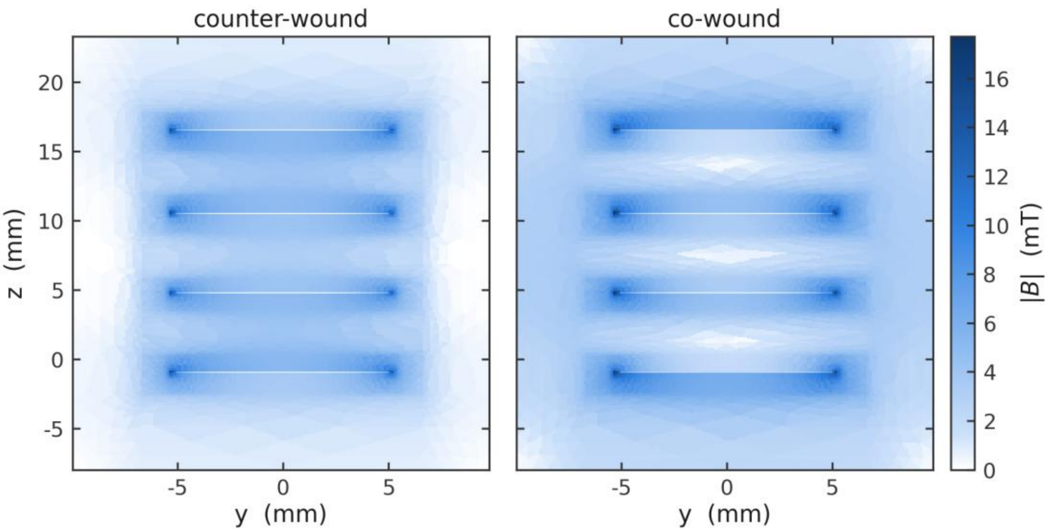

Figure 4.21 Flux density magnitude on the same cross-section, counter-wound against co-wound (rated operation).

Figures 4.20 and 4.21 compare the two winding directions at rated current. Counter-winding reduces the equivalent inductance and stored magnetic energy of the 50 mm segment by a factor of approximately 2.8, while redistributing flux towards the tapes. The outside-envelope share of the B^2 integral falls from 81.7% to 33.3%, whereas the inside-envelope integral rises by about 31%. Both effects arise from the same redistribution, linking gap selection to cooling and AC loss.

5. Summary

5.1 Model and computed cases

A three-dimensional electromagnetic, thermal and external-circuit model has been developed for a series-connected counter-wound, copper-stabilised REBCO resistive SFCL. The representative segment is linked to device quantities by a single length factor, and the fault, interruption and rated-current cases use the same geometry, constitutive laws and solution framework.

5.2 Electrical response

The SFCL develops most of its limiting impedance during the first fault half cycle. Later current peaks decline as the conductor resistance increases. The fault-inception phase changes the peak-current reduction by several percentage points and should accompany any reported peak metric. The terminal-voltage maximum follows the current maximum after a short delay, and breaker opening leaves only the residual current associated with the finite open-state resistance. Under the present periodic weak-zone pattern, the uniform-Jc control changes the first peak by less than 1%, indicating limited sensitivity of the device-level current in this operating case.

5.3 Thermal response

During the fault, the superconducting-layer hot spot separates progressively from the solid average temperature until interruption arrests the rising envelope. Before the boiling-transition branch is reached, temperature rise increases monotonically with clearing time, while the liquid-nitrogen boundary governs post-interruption cooling. A temperature threshold describes thermal decay. Rated-current capability requires a loaded recovery calculation. The prescribed weak-zone pattern determines the location and magnitude of the computed thermal maximum; the uniform-Jc control remains nearly uniform in the axial direction over the first 20 ms.

5.4 AC loss at rated current

At rated current, the superconducting layer accounts for nearly all of the AC loss. Counter-winding reduces net loop flux linkage and inductance but concentrates magnetic field within the tape envelope. Gap size, perpendicular field and loss per unit length must therefore be assessed together.

5.5 Energy balance

For the base case, cumulative Joule heat is consistent with the sum of surface cooling and retained enthalpy. A small electrical-port residual remains, but it is well below the changes associated with fault limitation and conductor heating.

6. Discussion and Further Work

6.1 Calibrating the liquid-nitrogen boundary for this geometry

The heat-transfer boundary used in the model is derived from horizontal upward-facing plates, unlike the vertical narrow-gap stack considered here. Geometry-specific calibration requires both pool-boiling measurements on the stack and a transient heat-transfer model resolved on the millisecond timescale.

The characteristic time for natural convection to develop in liquid nitrogen is of the same order as the simulated fault duration (Balakin et al., 2017; Delov et al., 2020). Under high heat flux, vapour generation can exceed buoyant removal, forming gas columns or an intermittent vapour film in the vertical gap and restricting lateral liquid supply. Experiments in narrow slots report both reduced critical heat flux and confined boiling patterns (Fujita et al., 1988; Bonjour and Lallemand, 1998), while coated-conductor tests associate delayed rewetting with bubble retention (Zhi et al., 2020; Luo et al., 2022a). Steady boiling curves omit the short-time development of convection and vapour structures. Thus, calibration should use heat-flux steps on the timescale of the fault (Duluc, Stutz and Lallemand, 2008; Delov et al., 2020).

Gap width affects vapour removal, dielectric withstand and counter-winding coupling. A wider channel improves liquid replenishment and venting but weakens flux-linkage cancellation and increases AC loss. A narrower channel promotes bubble retention and may reduce two-phase insulation strength (Song et al., 2018; Sauers et al., 2011; Hayakawa et al., 2015). Quantitative design therefore requires a joint parameter sweep of heat transfer, inductance and loss using geometry-matched boiling and dielectric data.

6.2 Near-field flux concentration and the loss trade-off in the counter-wound configuration

Opposing currents in adjacent tapes reduce the far-field magnetic vector potential, net loop flux linkage, stored magnetic energy and equivalent inductance (Clem, 2008; Nguyen, Ashworth and Willis, 2009; Song et al., 2018). For the 50 mm segment, the calculated inductance is lower by a factor of approximately 2.8.

Counter-winding moves magnetic energy from the far field towards the tapes. The outside-envelope share of the B^2 volume integral decreases from 82% for co-winding to 33% for counter-winding. Beyond twice the tape envelope, the share falls from 41% to 6.5%, while the inside-envelope integral rises by approximately 31%. This concentration is consistent with the calculated edge and surface losses of closely spaced non-inductive coils (Grilli et al., 2010).

The loss response is anisotropic. The component $B_{\perp}$ normal to the tape face governs flux penetration and hysteresis loss, while reducing local J_c through the anisotropic critical-current relation (Brandt and Indenbom, 1993; Norris, 1970; Pardo and Grilli, 2012; Amemiya et al., 2005). Increased $B_{\perp}$ at the tape edges can increase local electric field and Joule dissipation. A reduction in net loop flux linkage therefore need not produce a proportional reduction in AC loss.

Experiments and refined calculations should separate losses associated with $B_{\perp}$ and $B_{\parallel}$ and compare the effect of winding direction with transport-loss measurements on variable-gap bifilar samples (Song et al., 2018).

6.3 Establishing an independent cross-comparison basis

The numerical checks reported here test internal consistency. Independent validation would require either measurements on a stack of matching construction or a solution of the same case with a different formulation. Suitable comparison quantities are port voltage, line current, local temperature and cumulative energy.

6.4 Loaded recovery and reclosing extensions

Recovery comprises thermal relaxation followed by restoration of rated-current capability. The calculated cases describe the temperature evolution after interruption. A reclosing study should restore the healthy load after fault clearance and determine the waiting time from both current-carrying capability and terminal voltage drop.

Artificial Intelligence (AI) Tools Usage Declaration

In the preparation of this thesis, the following AI tools were used:

1. ChatGPT / Claude: These tools were used to assist with brainstorming, initial structuring of ideas, and polishing the academic language to enhance the clarity and flow of the text.

2. Grammarly Premium: Used for spelling, grammar and punctuation checks.

All generated suggestions were critically reviewed, evaluated, and revised by the author. The research design, data analysis, scientific conclusions, and the core intellectual content presented in this thesis remain entirely the author's original work.

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